CO2xNxPhrag: Porewater Methane

July 2026 Samples

2025-08-22

Contents

##Run Information

```r
cat("Run Information: Brenden D. Blakley ") #lets you know what section you're in
```

```
## Run Information: Brenden D. Blakley
```

```r
library(here)
```

```
## here() starts at C:/Users/blakleybd/OneDrive - Smithsonian Institution/Documents/GitHub/GCREW
```

```r
#set the run date & user name
sample_year <- "2026"
sample_month <- "July"
user <- "Brenden Blakley"
#identify the files you want to read in
#read in as a list to accommodate multiple runs in a month

GHG_files <- c("Raw Data/20260817_GCReW_Phrag_GENX_formatted.csv" )


# Define the file path for QAQC log file - NO Need to change just check year
log_path <- here("Porewater QAQC Logs", "GCReW_CH4_QAQC_Log.csv")

# Define final path for data be sure to change year and month and run number
final_path <- "Processed Data/GCReW_CO2xNxPhrag_Porewater_CH4_202607.csv"

#Gas in Water Calculations
# "TempC" column value in Celsius (equilibration temperature) and volume of gas (in liters) in syringe
TempC = 25 #Equilibration Temperature (Celsius)

volume_g = 0.015 #volume of gas (in liters) in syringe when equilibrized
volume_w = 0.015 #volume of water (in liters) in syringe when equilibrized (should be equal to volume_g

#record any notes about the run or anything other info here:
run_notes <- ""



#Collection_Dates = "Raw Data/Sample_Collection_Dates_2024.csv"

cat(run_notes)
```

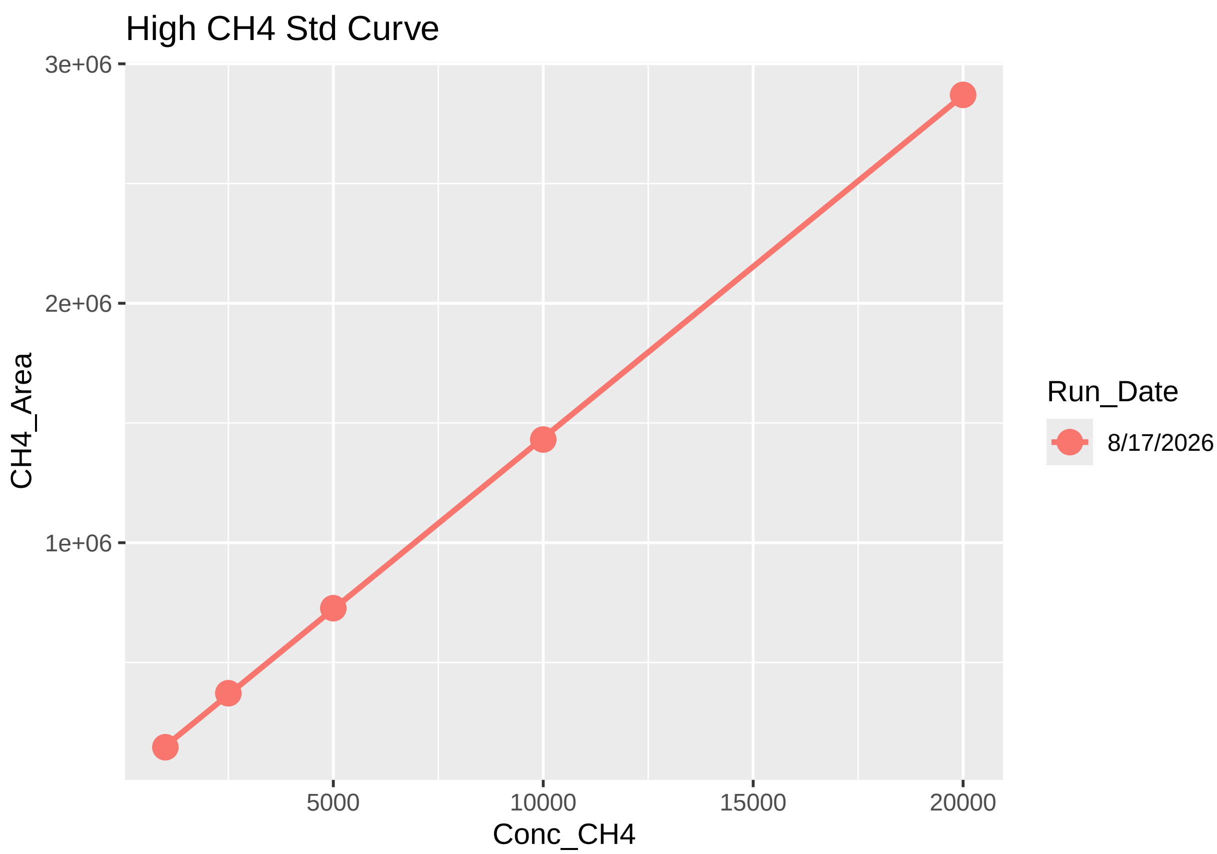
##Set Up

##Read in metadata and create similar sample IDs for matching to samples

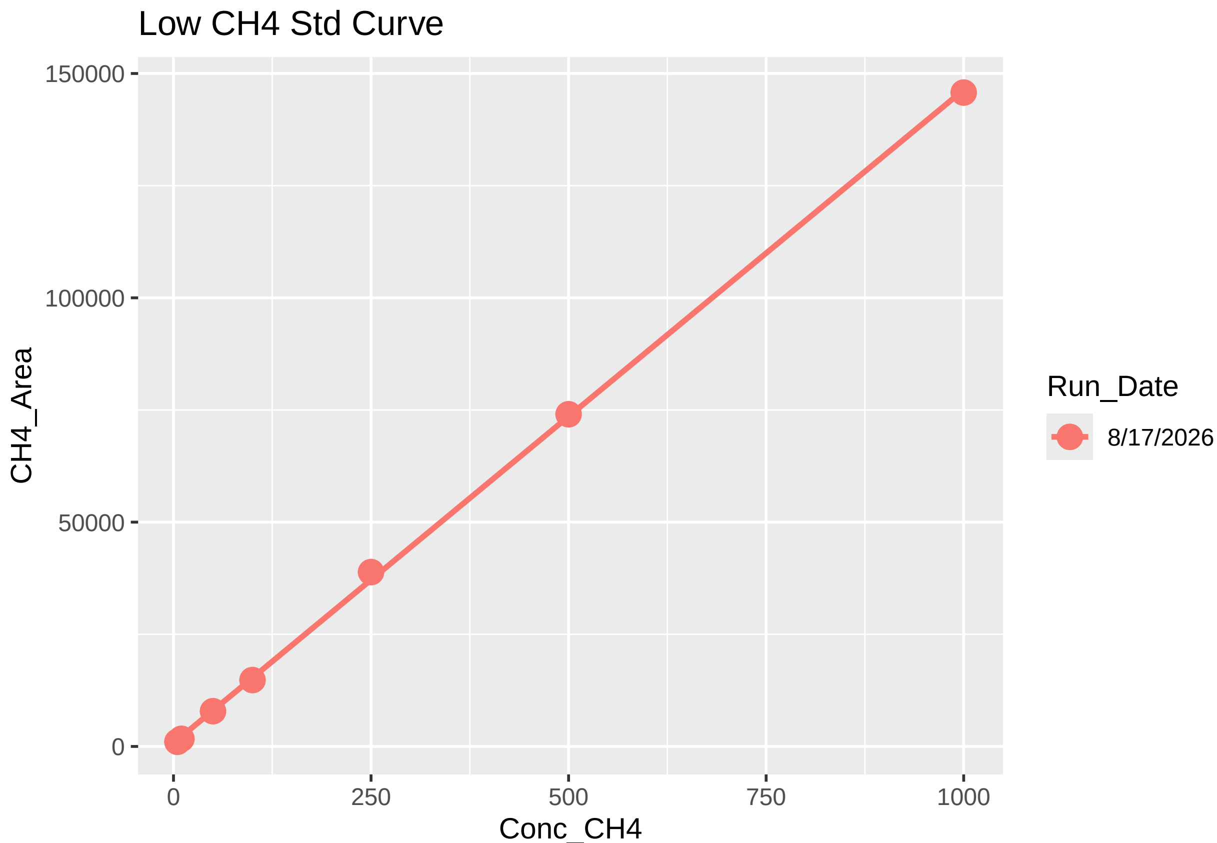
1 Read in data files

2 Assess standard curves

```
## `geom_smooth()` using formula = 'y ~ x'
```




```
## `geom_smooth()` using formula = 'y ~ x'
```

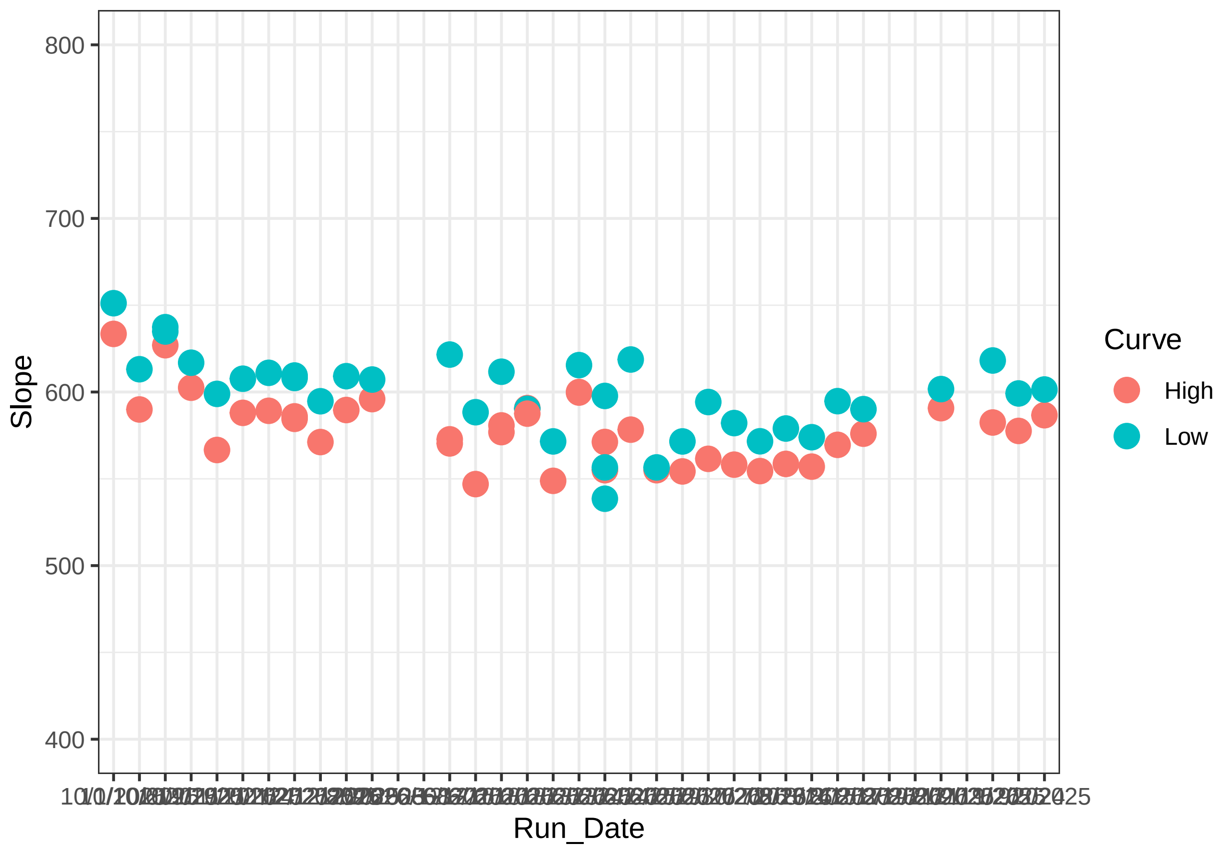



```
## Joining with `by = join_by(Run_Date)`

##   Curve        R2    Slope    Intercept  Run_Date
## 1  High 0.9986295 576.8990  173986.2174 6/16/2025
## 2   Low 0.9990260 611.6961   -3791.2684 6/16/2025
## 3  High 0.9999101 570.3925  138747.3478 6/12/2025
## 4   Low 0.9979098 621.5734    3224.2535 6/12/2025
## 5  High 0.9989882 590.7374 -109278.7391 6/18/2025
## 6   Low 0.9990995 590.3138     954.7682 6/18/2025

## Warning: There was 1 warning in `mutate()`.
## i In argument: `Run_Date = mdy(Run_Date)`.
## Caused by warning:
## !  6 failed to parse.

## Warning: Removed 14 rows containing missing values or values outside the scale range
## (`geom_point()`).
```



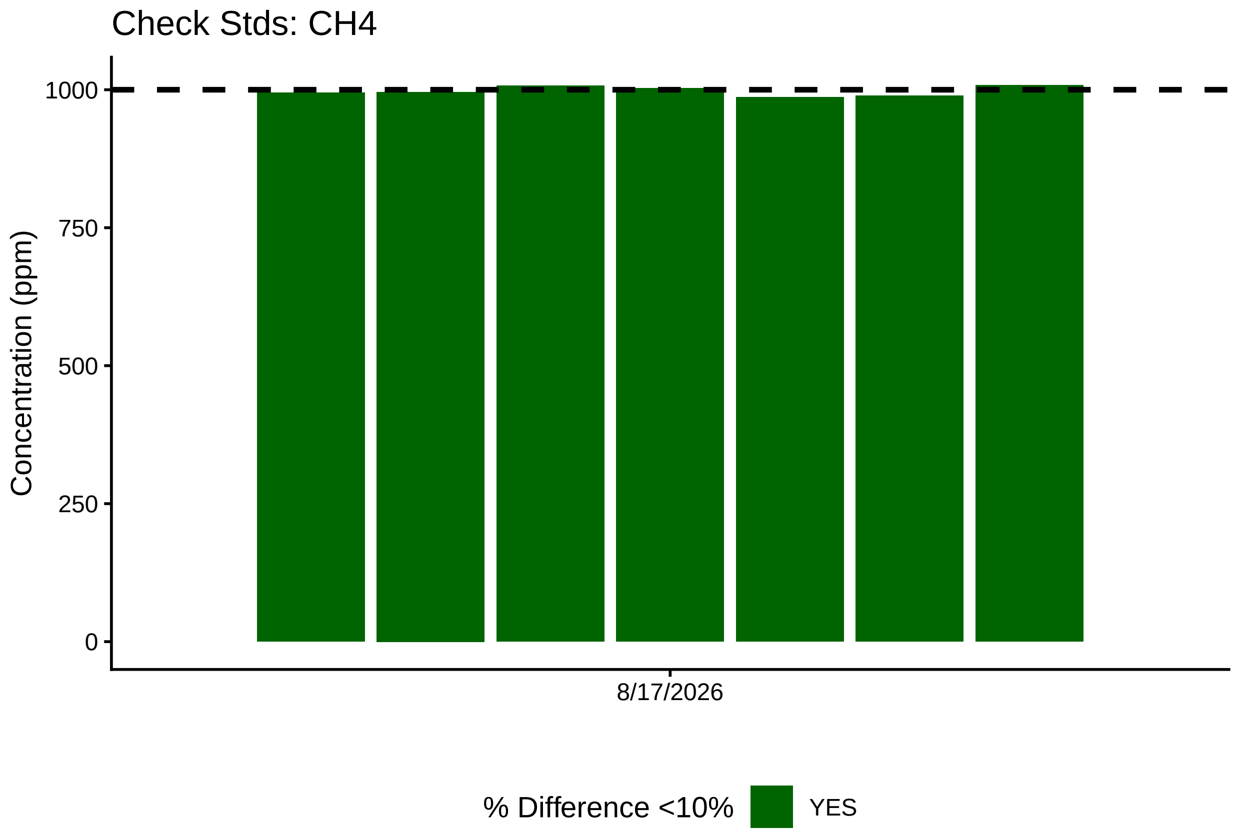

```
## [1] "High_R2 is Good-8/17/2026"
```

```
## [1] "Low_R2 is Good-8/17/2026"
```

2.1 Now calculate the CH4 & CO2 concentrations in ppm

3 Anlayze the Check Standards

```
## >60% of CH4 Check Standards are within range of expected concentration - PROCEED
```



4 Analyze the Blanks

```
## Assess Blanks

## >60% of CH4 Blanks are below the lower 25% quartile of samples - PROCEED
```

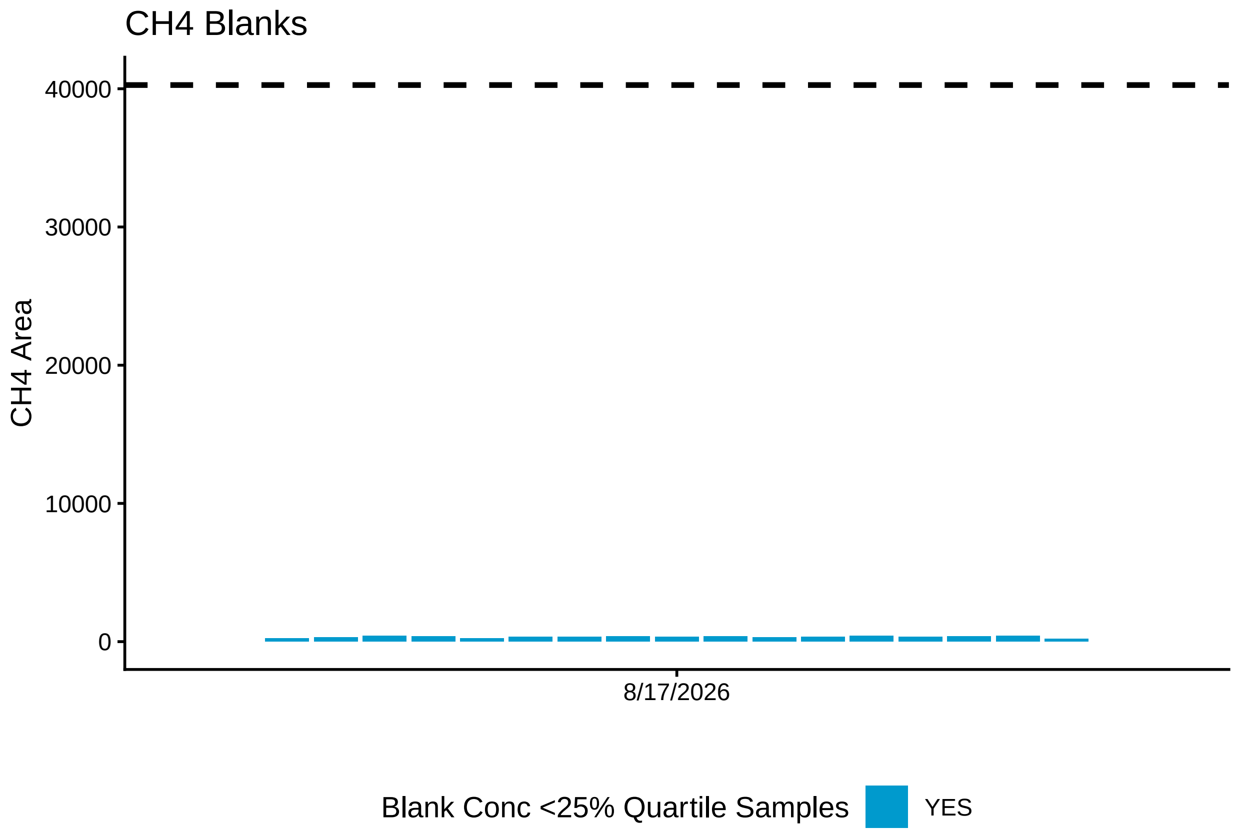
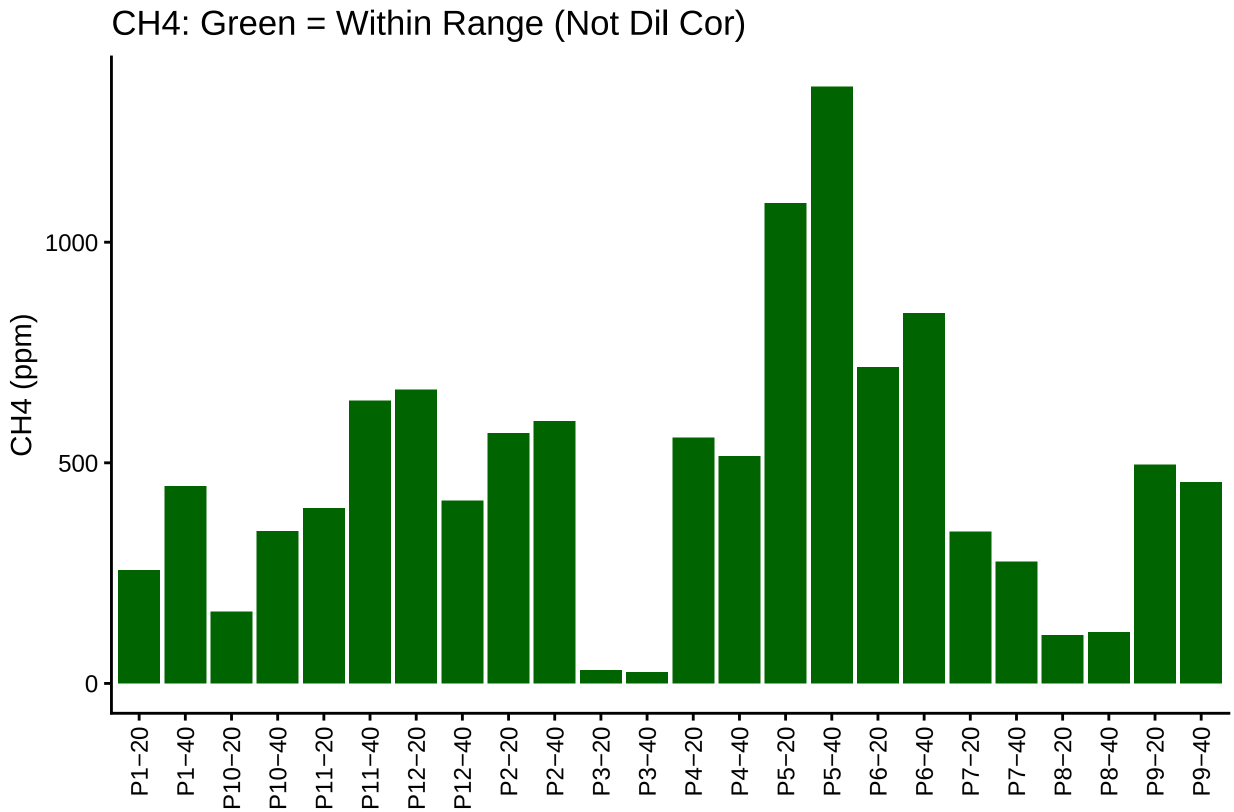
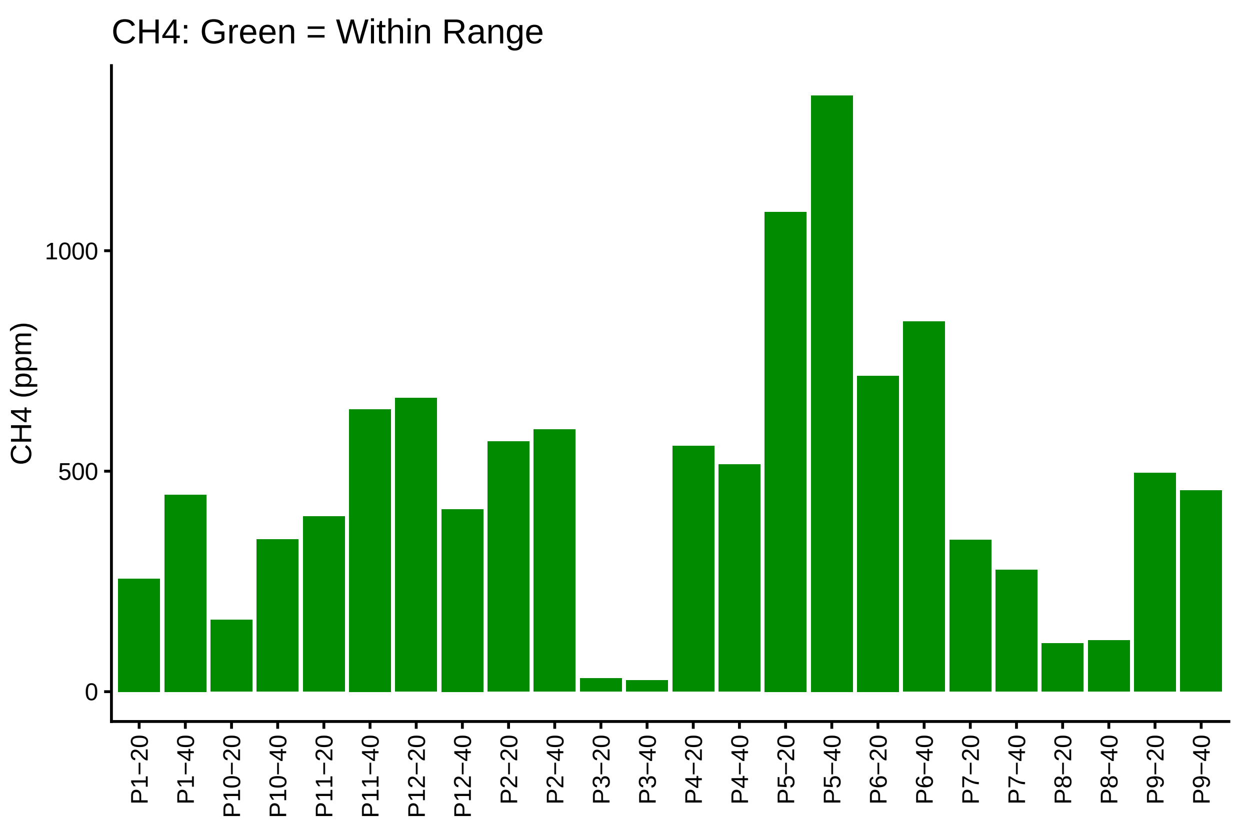


Table 1: Mean Area of Blanks

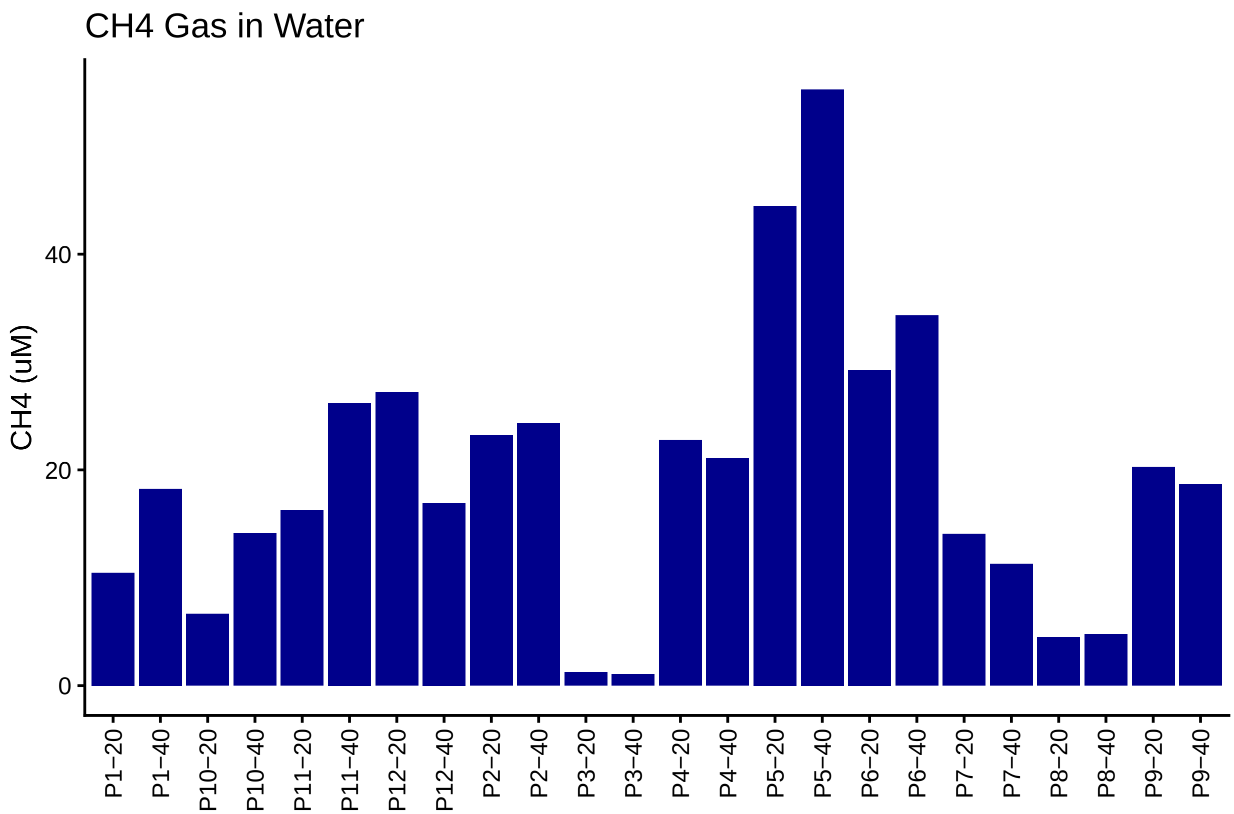
Test	Blank_Mean_Area
CH4	342.1176

5 Dilution correct samples and Check Range





6 Calculate gas in water



7 Merge samples with Metadata & check all samples are present

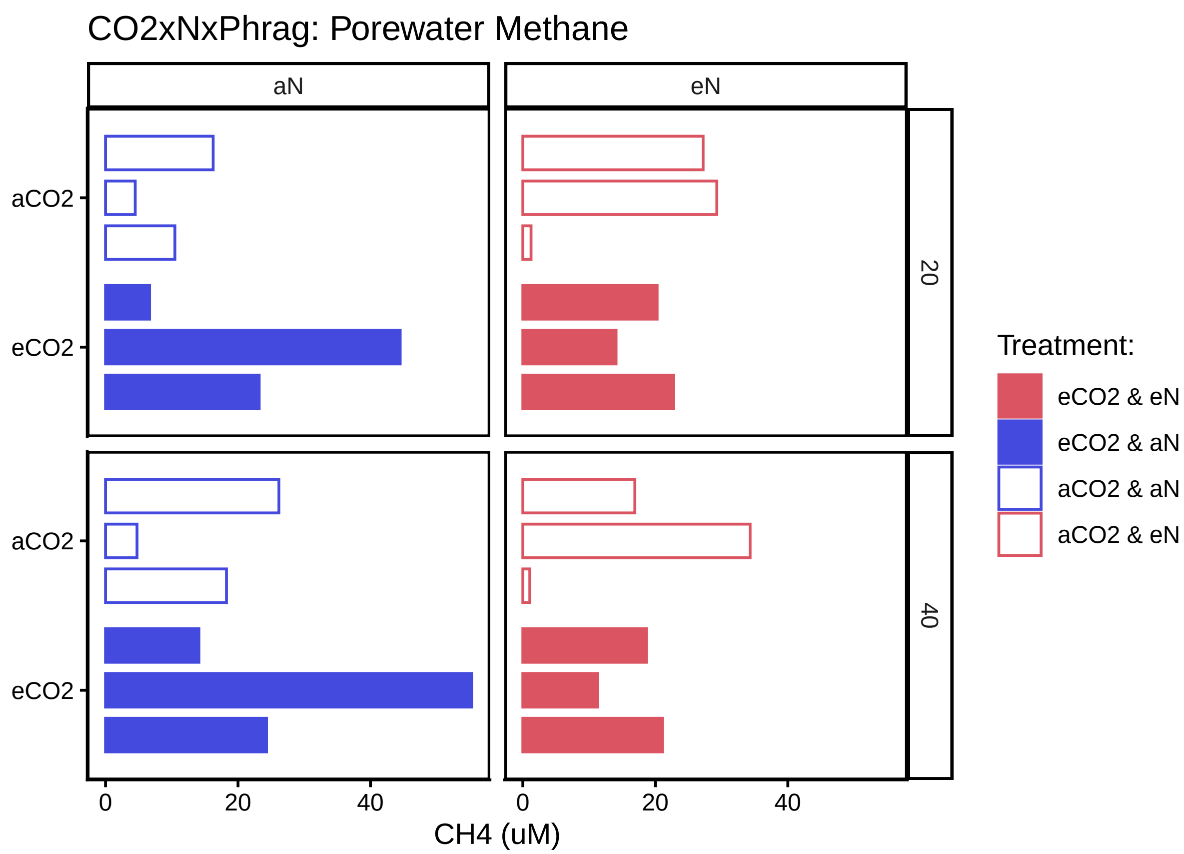
```
## Merge Samples with Metadata
```

```
## All sample IDs are present in data
```

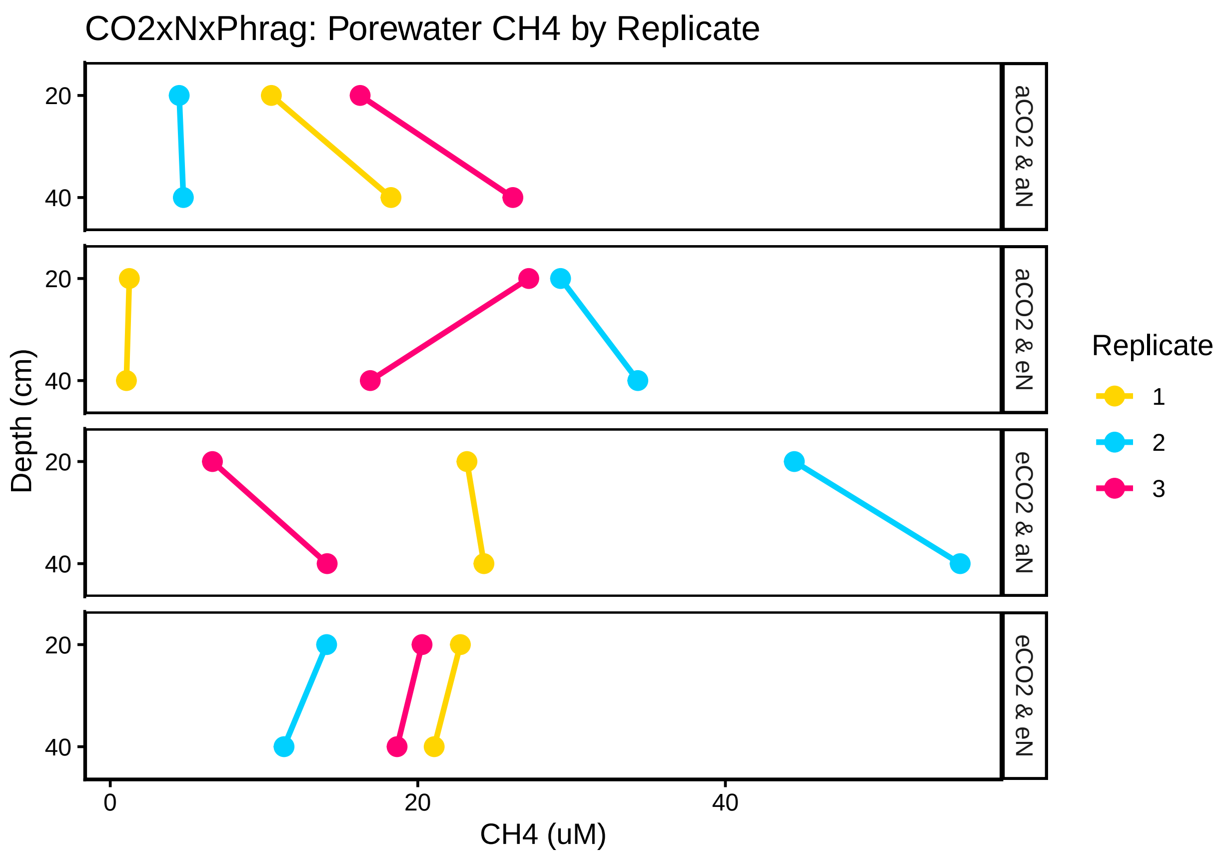
7.1 Organize Data

7.2 Visualize Data

```
## Visualize Data
```



CO2xNxPhrag: Porewater Methane
aN
eN
aCO2
eCO2
aCO2
eCO2
20
40
0
20
40
0
20
40
CH4 (uM)
Treatment:
eCO2 & eN
eCO2 & aN
aCO2 & aN
aCO2 & eN



#Export Processed Data

```
write.csv(All_Clean_Data, final_path)
```

#end